UNIVERSITY OF AUCKLAND
Waipapa Taumata Rau
NEW ZEALAND

SCIENCE

COMPSCI 714
Lectutorial 5: Learning with images - CNNs & ViT

How to implement and use pretrained CNNs + Principles and use of the Vision Transformer

Instructor: Thomas Lacombe

Learning outcomes

- Implement and use CNNs with PyTorch
- Understand the principles of the Vision Transformer (ViT) and how to use it with PyTorch

Extra material (optional)

- Understand 1D convolution and how it can be applied to sequential data.
- Understand the application of 1D convolution in the WaveNet architecture.

Extra resources

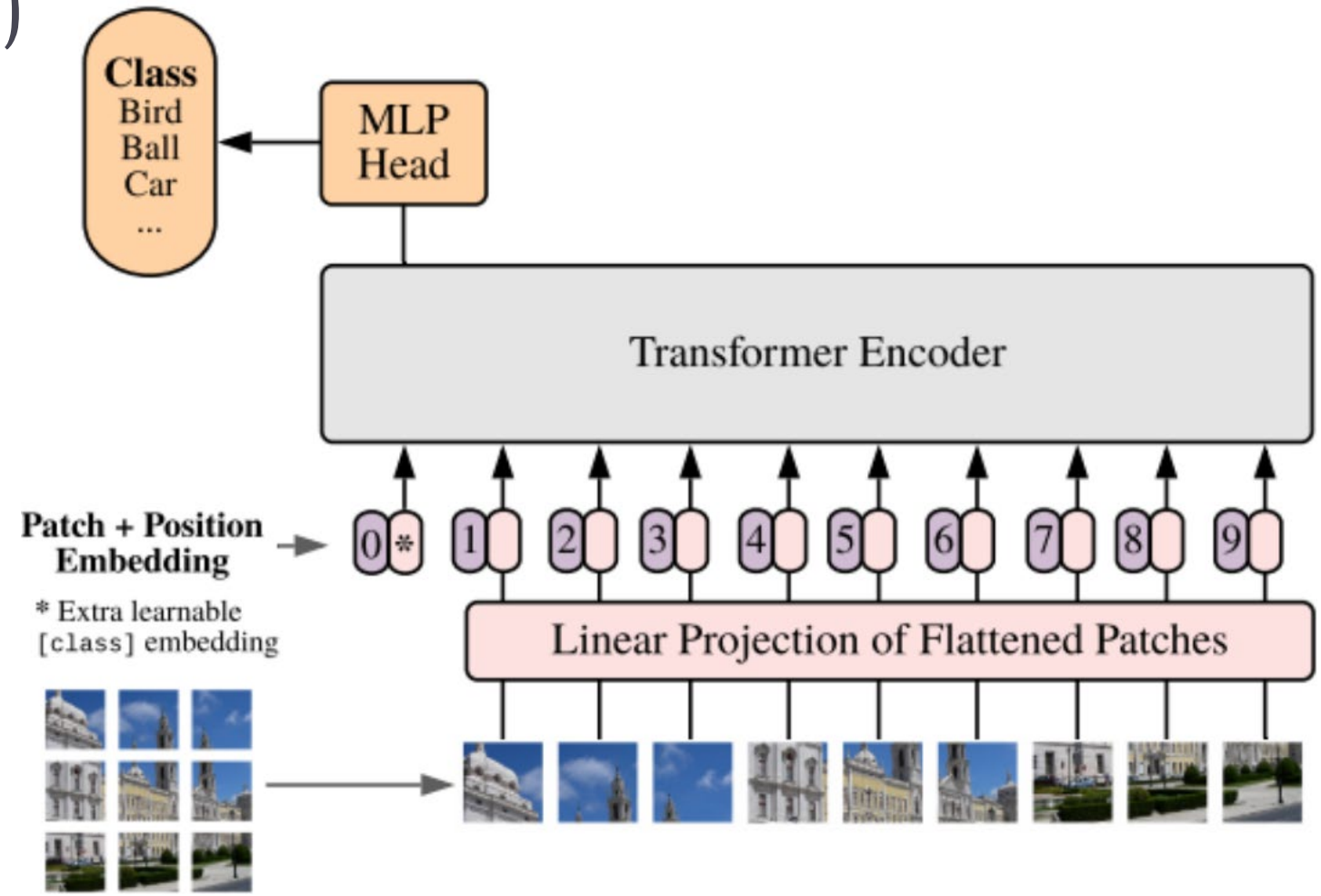
If you want to go further:

- *Hands-on Machine Learning with Scikit-Learn, Keras and Tensorflow*, Géron A.
  - Chapter 15 and 16.
- *Deep Learning*, Goodfellow I. et al.
  - Part II Deep Networks, chap. 9.

Coding time 1: Convolutional Neural Networks (CNNs)

Vision Transformer (ViT)

- Proposed by Google Brain team in 2020.
- Chops the image into 16x16 squares/patches (flattened).
- Treat the sequence of squares as if it was a sequence of word embeddings.
- Position embedding.
- Use the transformer encoder to generate a representation that is fed to a MLP head for classification.



[Vision Transformer, 2020]
https://blog.research.google/2020/12/transformers-for-image-recognition-at.html

Vision Transformer (ViT)

- The ViT was able to beat SoTA CNN models on the ImageNet image classification task (millions of images from 1000 different classes).

- However, it needed more training data, which makes sense as it has less *inductive biases* than a CNN when working with images.
  - *Inductive biases* refer to implicit assumptions made by a model due to its architecture (e.g., CNN implicitly assumes that patterns learned at one location will likely be useful in another locations, because of the use of covolutional layers).

- Since 2020, a lot of developments happened with ViT, see “Hands on ML, Ed3”, chap. 16, and survey “A Survey on Visual Transformer” (2023).

Coding time 2: ViT VS ResNet using Hugging Face

Extra material slides (not evaluated)

Convolutions for Sequential Data

- 1 D Convolution
- WaveNet
- (See extra notebook for implementation with TensorFlow/Keras)

1D Convolution

Output	4											

Filter (1x1)	2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6										

Filter (1x1)

2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6	14									
Filter (1x1)			2									
Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6	14	2	14	4	9	10	2	5	8	2

Filter (1x1)

2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	**35**											
Filter (1x3)	3	5	2									
Input	**2**	**3**	**7**	**1**	**7**	**4**	**9**	**10**	**2**	**5**	**8**	**2**

1D Convolution

Output	**35**	**46**										
Filter (1x3)		3	5	2								
Input	**2**	**3**	**7**	**1**	**7**	**4**	**9**	**10**	**2**	**5**	**8**	**2**

1D Convolution

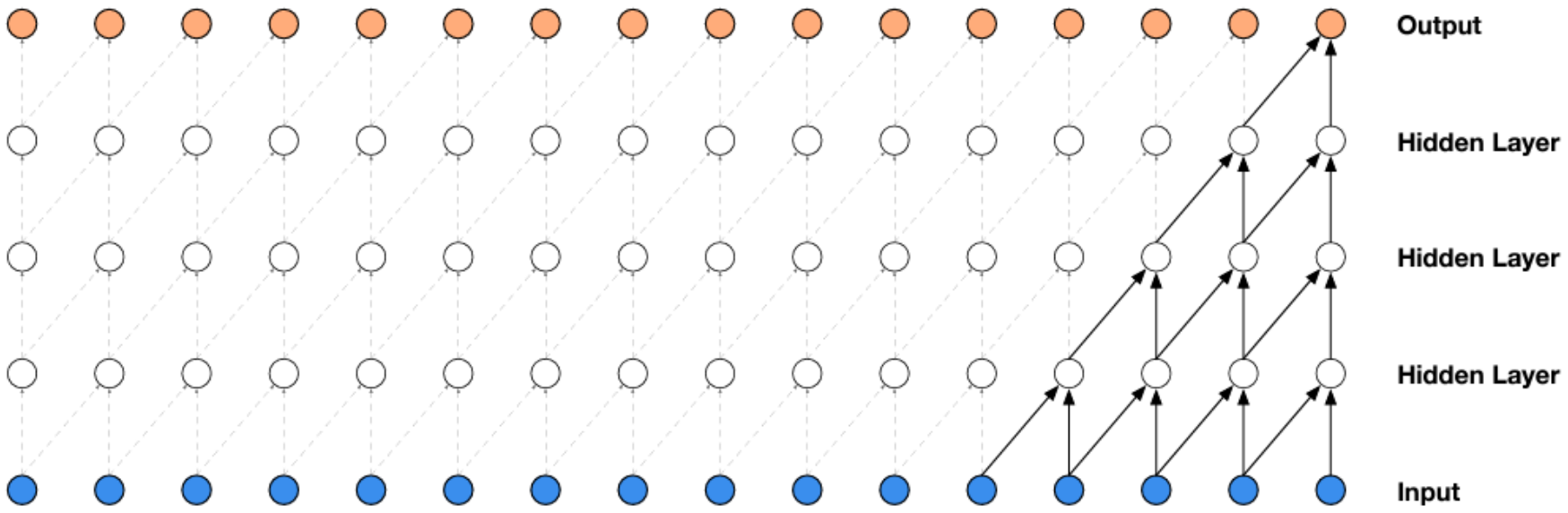
Output	35	46	40	47	59	67	81	50	47	59		
Filter (1x3)										3	5	2
Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

- When is it useful?
  - **Train 1D convolution layer on sequential data**: each kernel in the layer will learn a single short sequential pattern (no longer than the kernel size)
  - Can be used to learn features on the sequence, but also to downsample the sequence (if valid padding or stride > 1).
  - Can be combined with RNN layers (see Tutorial 5).

WaveNet

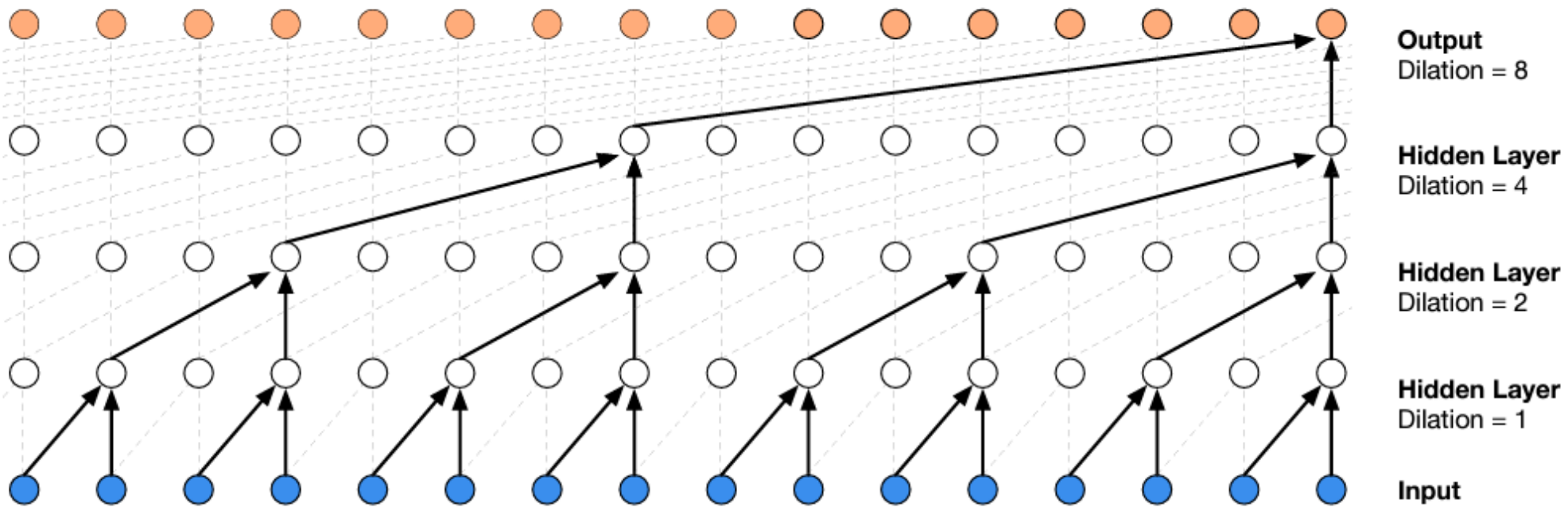
- Network proposed in 2016 by DeepMind/Google for audio waveform generation
  - Stack 1D causal convolution layers
  - Causal convolution means that a neuron is connected to current and previous time steps (not future).



[WaveNet, 2016]
https://deepmind.google/discover/blog/wavenet-a-generative-model-for-raw-audio/

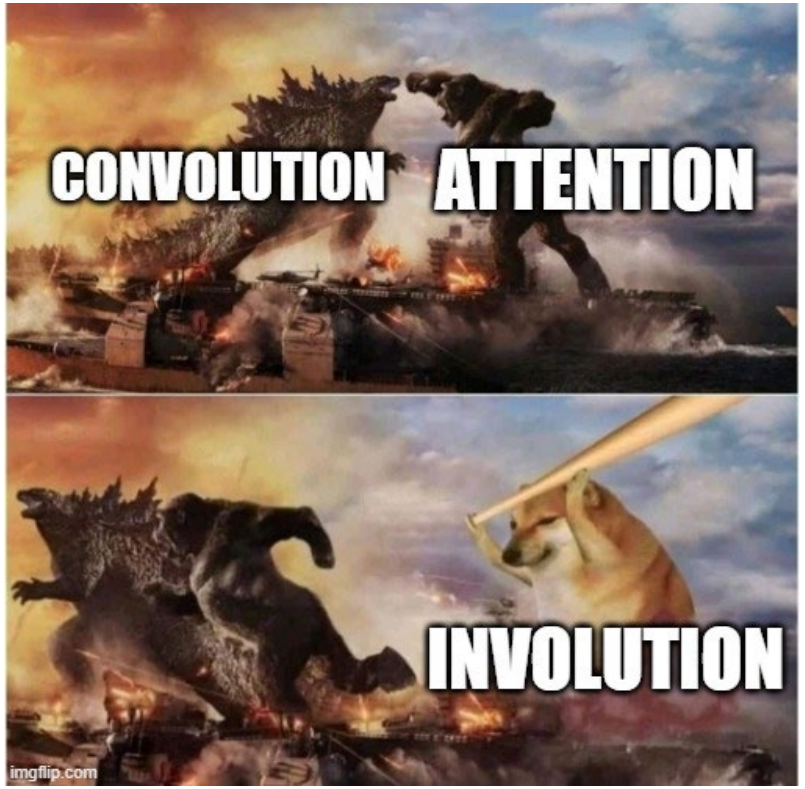
Wavenet

- Causal dilated convolution
  - First hidden layer sees 2 input time steps, next one sees 4, the next one sees 8, etc.
  - Lower layers learn short term patterns while deeper layers learn long-term patterns.



[Wavenet, 2016]
https://deepmind.google/discover/blog/wavenet-a-generative-model-for-raw-audio/

Conclusion meme



https://keras.io/examples/vision/involution/